

JINGDONG ZHANG

Ph.D. Student in Computer Science, Texas A&M University

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RESEARCH INTERESTS

My research builds scalable foundation models for spatial intelligence, with interests in:

- ♦ **Spatial reasoning & scene understanding.** Perceiving and reasoning about the world from visual observations, toward embodied spatial intelligence.
- ♦ **Multi-modal learning.** Vision-language models and agentic systems that ground, search, and reason across modalities.
- ♦ **Generative models.** Controllable visual editing and content creation.

EDUCATION

Texas A&M University

Ph.D. in Computer Science, Department of Computer Science and Engineering

Advisors: Prof. Wenping Wang and Prof. Xin Li

Aug. 2023 – Present

College Station, TX

Fudan University

B.Eng. in Intelligent Science and Technology (Excellent Class), School of Information Science and Engineering

Sep. 2019 – Jun. 2023

Shanghai, China

INDUSTRY EXPERIENCE

NVIDIA Research

Research Intern, mentored by Chris Choy

- ♦ Foundational 3D/4D understanding: S4VY, a Segment Anything model built on feed-forward 4D visual geometry, with agentic natural-language grounding.

May 2026 – Aug. 2026

Adobe

Research Intern, mentored by Xiaoahang Zhan and Lingzhi Zhang

- ♦ Generative soft inpainting: UniSER, a foundation model that removes diverse soft effects (lens flare, haze, shadows, reflections) in one framework. Accepted by CVPR 2026.

May 2025 – Aug. 2025

Tencent America

Research Intern, mentored by Weikai Chen and Bo Yang

- ♦ High-quality 3D asset generation: SPGen, which leverages 2D diffusion priors through consistent Spherical Projection maps for single-image 3D shape generation. Accepted by SIGGRAPH Asia 2025.

May 2024 – Aug. 2024

RESEARCH EXPERIENCE

Aggie Graphics Group, Texas A&M University

Ph.D. Student, advised by Prof. Wenping Wang and Prof. Xin Li

- ♦ Panoramic scene understanding: MTPano, a multi-task panoramic foundation model trained label-free from perspective dense priors. Accepted by SIGGRAPH 2026.
- ♦ Embodied visual search: Imagining in 360, which decouples spatial imagination from action planning for humanoid visual search. Accepted by NeurIPS 2026.

Aug. 2023 – Present

Hong Kong University of Science and Technology

Research Assistant, advised by Prof. Dan Xu

- ♦ Multi-task learning with partially annotated data: hierarchical task tokens for cross-task interaction and pseudo-label discovery in scene parsing. Accepted by ACM MM 2025.

Feb. 2022 – Nov. 2023

Embedded Deep Learning and Visual Analysis Lab, Fudan University

Research Assistant, advised by Prof. Tao Chen and Prof. Jiayuan Fan

- ♦ Multi-task dense prediction: BridgeNet, which builds bridge features as comprehensive intermediate multi-task representations. Accepted by TPAMI.
- ♦ Cross-domain pedestrian detection: background-focused distribution alignment for domain-adaptive one-stage detectors. Accepted by TIP.

Jul. 2021 – Aug. 2023

PUBLICATIONS

[1] S4VY: Segment Anything in Feed-Forward 4D Visual Geometry

Jingdong Zhang, Xin Li, Jan Kautz, Wenping Wang, Chris Choy

Under review, 2026

- [2] **Beyond Thinking: Imagining in 360 for Humanoid Visual Search**
Jingdong Zhang, Yizhou Wang, Zhengzhong Tu, Xin Li, Wenping Wang, Xiaohang Zhan
NeurIPS 2026 [arXiv]
- [3] **MTPano: Multi-Task Panoramic Scene Understanding via Label-Free Integration of Dense Prediction Priors**
Jingdong Zhang, Xiaohang Zhan, Lingzhi Zhang, Yizhou Wang, Zhengming Yu, Jionghao Wang, Wenping Wang, Xin Li
SIGGRAPH 2026 [arXiv] [Project] [Code] [Demo]
- [4] **Promoting LDR Panoramas to HDRI with Rendering-Space Losses**
Zhengming Yu, Li Ma, Mingming He, **Jingdong Zhang**, Jionghao Wang, Ning Yu, Paul Debevec
SIGGRAPH 2026 Poster [Paper] [Project] [Code]
- [5] **UniSER: A Foundation Model for Unified Soft Effects Removal**
Jingdong Zhang, Lingzhi Zhang, Qing Liu, MangTik Chiu, Connelly Barnes, Yizhou Wang, Haoran You, Xiaoyang Liu, Yuqian Zhou, Zhe Lin, Eli Shechtman, Sohrab Amirghodsi, Xin Li, Wenping Wang, Xiaohang Zhan
CVPR 2026 [Paper] [arXiv] [Project] [Code] [Dataset]
- [6] **SPGen: Spherical Projection as Consistent and Flexible Representation for Single Image 3D Shape Generation**
Jingdong Zhang, Weikai Chen, Yuan Liu, Jionghao Wang, Zhengming Yu, Zhuowen Shen, Bo Yang, Wenping Wang, Xin Li
SIGGRAPH Asia 2025 [Paper] [arXiv] [Project]
- [7] **SolidGS: Consolidating Gaussian Surfel Splatting for Sparse-View Surface Reconstruction**
Zhuowen Shen, Yuan Liu, Zhang Chen, Zhong Li, Jiepeng Wang, Yongqing Liang, Zhengming Yu, **Jingdong Zhang**, Yi Xu, Scott Schaefer, Xin Li, Wenping Wang
arXiv 2024 [arXiv] [Project]
- [8] **Multi-Task Label Discovery via Hierarchical Task Tokens for Partially Annotated Dense Predictions**
Jingdong Zhang, Hanrong Ye, Xin Li, Wenping Wang, Dan Xu
ACM MM 2025 [Paper] [arXiv] [Code]
- [9] **BridgeNet: Comprehensive and Effective Feature Interactions via Bridge Feature for Multi-task Dense Predictions**
Jingdong Zhang, Jiayuan Fan, Peng Ye, Bo Zhang, Hancheng Ye, Baopu Li, Yancheng Cai, Tao Chen
IEEE TPAMI 2025 [Paper] [IEEE] [Code]
- [10] **Rethinking Cross-Domain Pedestrian Detection: A Background-Focused Distribution Alignment Framework for Instance-Free One-Stage Detectors**
Yancheng Cai, Bo Zhang, Baopu Li, Tao Chen, Hongliang Yan, **Jingdong Zhang**, Jiahao Xu
IEEE TIP 2023 [IEEE] [Code]

ACADEMIC SERVICE & TEACHING

Reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICRA, AAAI, TPAMI, TVCG

Teaching Assistant, Texas A&M University: CSCE 633 Machine Learning; CSCE 489 Special Topics in Computer Science and Engineering; CSCE 442 Scientific Programming; CSCE 222 Discrete Structures for Computing

SELECTED AWARDS

First prize, Advanced Driving Assistance System (ADAS) National Competition, Dell Corporation	2021
Third prize, Outstanding Undergraduate Student Scholarship, Fudan University	2021–2022
Second prize, Outstanding Undergraduate Student Scholarship, Fudan University	2019–2020
Outstanding Student, Fudan University	2019–2020

PROJECTS

ADAS National Competition (Team Leader)	May 2021
♦ Built a pipeline combining real-time detection and segmentation for perception, followed by planning and control; won the first prize. [Video]	

SKILLS

Programming: Python, C, MATLAB ♦ **Deep learning:** PyTorch